Hui Guo

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Research Interests

My research is broadly in **statistical machine learning for imperfect data**. Methodologically, I am interested in learning with imperfect supervision, noisy label learning, weak and crowdsourced supervision, robust and distributionally robust learning, and uncertainty quantification. In an applied context, I am interested in developing models and algorithms for medical AI and foundation models for healthcare.

Education

- **Ph.D. in Computer Science, Western University** 09/2021–Present
  Department of Computer Science
  Advisors: Prof. Grace Y. Yi and Prof. Boyu Wang
  Dissertation focus: **Learning with Imperfect Data: Statistical and Machine Learning Methods**
  Expected completion: Summer 2026 | Graduate course average: 95.75%

- **B.Sc. in Mathematics and Applied Mathematics, Nankai University** 09/2016–06/2021
  Thesis: A Generalization on Conditional Distortion Risk Measures

- **B.Econ. in Finance, Nankai University** 09/2016–06/2021
  Thesis: Pairs Trading Strategy Based on Hurst Exponent
  Undergraduate course average: 90.30%

Publications

Preprints and Submissions

- Dual-Granularity Learning for Regression with Continuous Noisy Labels
  Hui Guo, Boyu Wang, Grace Y. Yi
  *Submitted to **Neural Information Processing Systems (NeurIPS)***, 2026.

Journal Publications

- Addressing both Variable Selection and Misclassified Responses with Parametric and Semiparametric Methods
  Hui Guo, Grace Y. Yi, Boyu Wang
  Bernoulli 32(2): 1303–1327, 2026. [paper] [code] [arXiv]

- Physician Effects in Critical Care: A Causal Inference Approach Through Propensity Weighting with Parametric and Super Learning Methods
  Yuan Bian, Yu Shi, **Hui Guo**, Grace Y. Yi, Wenqing He
  Journal of Data Science 23(1): 130–148, 2025. [paper]

Conference Publications

- Revisiting Source-Free Domain Adaptation: A New Perspective via Uncertainty Control
  Gezheng Xu*, **Hui Guo***, Li Yi, Charles Ling, Boyu Wang, Grace Y. Yi *(*Equal contribution)*
  International Conference on Learning Representations (ICLR), 2025. [paper] [code]

- Learning from Noisy Labels via Conditional Distributionally Robust Optimization
  Hui Guo, Grace Y. Yi, Boyu Wang
  Neural Information Processing Systems (NeurIPS), 2024. [paper] [code] [arXiv]

- Label Correction of Crowdsourced Noisy Annotations with an Instance-Dependent Noise Transition Model
  Hui Guo, Boyu Wang, Grace Y. Yi
  Neural Information Processing Systems (NeurIPS), 2023. [paper] [code]

Working Papers

- Instance-Dependent Bayesian Modeling and Robust Training with Crowdsourced Noisy Data
  Hui Guo, Grace Y. Yi
  Manuscript in preparation.

- Reliability-Aware Longitudinal Clinical Prediction from Multimodal Electronic Health Records
  Hui Guo, Yi Wan, Boyu Wang, Grace Y. Yi
  Manuscript in preparation.

Awards and Distinctions

- **Robert and Ruth Lumsden Graduate Fellowship in Science**, Western University 2024–2025
  Awarded to one PhD student per department in Science, selected based on academic excellence, research ability, and leadership, with criteria aligned with NSERC doctoral scholarship standards.

- **The Fourth Research Challenge Contest of Nankai University** May 2019
  Awarded honorable prize.

- **Students' Innovation and Entrepreneurship Training Program** 2018–2019
  Team was awarded the excellence award.

- **Nankai University Academic Excellence Scholarship** 2017–2018
  Awarded to the top 10% of students based on academic performance.

- **National Mathematical Modeling Contest for University Students** September 2017
  Team was awarded second prize in Tianjin.

- **Nankai University Talent Scholarship** 2016–2017
  Awarded to the top 10% of students based on academic performance.

Selected Talks

- Learning with Noisy Labels: Variable Selection and Misclassification Probability Modeling
  Lightning talk for Dr. Raymond Carroll's visit, Western University, June 2025.

- Revisiting Source-Free Domain Adaptation: A New Perspective via Uncertainty Control
  International Conference on Learning Representations (ICLR), Singapore, April 2025.

- Learning from Noisy Labels via Conditional Distributionally Robust Optimization
  Neural Information Processing Systems (NeurIPS), Vancouver, Canada, December 2024.

- Label Correction of Crowdsourced Noisy Annotations with an Instance-Dependent Noise Transition Model
  Neural Information Processing Systems (NeurIPS), New Orleans, USA, December 2023.

- Variable Selection for Logistic Regression Models with Misclassified Response
  Annual Meeting of the Statistical Society of Canada, Online, May 2022.

- Personalized Physician Recommendation for Critical Care Using the TreeSHAP Method (case study)
  Annual Meeting of the Statistical Society of Canada, Online, May 2022.

Workshops & Events

- Remarkable, Vector Institute
  Toronto, ON, February 2026

- Southern Ontario Learning Theory (SOLT) Workshop, Vector Institute
  University of Waterloo, Waterloo, ON, November 2025

Academic Service

- **Conference reviewer:** ICML 2026; ICLR 2024–2026 (3 years); AISTATS 2024–2026 (3 years); NeurIPS 2024–2026 (3 years).
- **Journal reviewer:** *Foundations and Trends in Signal Processing*; *Neural Networks*.

Selected Teaching Experience

Teaching Assistant, Western University London, ON, Canada

- **CS2209: Applied Logic for Computer Science** 2021–2026 (7 terms)
  An undergraduate core course for Computer Science students, with around 300 students, covering propositional and predicate logic, resolution, and logic programming in Prolog.
- **CS2035: Data Analysis and Visualization** 2022–2024 (3 terms)
  An undergraduate elective for Computer Science students, with around 80 students, covering statistical data analysis and visualization techniques using MATLAB.

Technical Skills

- **Programming:** Python, R, MATLAB, C++, LaTeX, Prolog.
- **Languages:** English (fluent), Mandarin Chinese (native).